

MET CS662 - Assignment #2

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1. Let $L = \{ab, aa, baa\}$. Which of the following strings are in L^* .
abaabaaabaa, *aaaabaaaa*, *baaaaabaaaab*, *baaaaabaa*?
abaabaaabaa $\in L^*$: *ab aa baa ab aa*
aaaabaaaa $\in L^*$: *aa aa baa aa*
baaaaabaaaab $\notin L^*$
baaaaabaa $\in L^*$: *baa aa ab aa*
2. Find grammars for $\Sigma = \{a, b\}$ that generate the set of:

- a. All strings with exactly one a .

$$S \rightarrow AaA$$

$$A \rightarrow bA \mid \lambda$$

This grammar can generate all strings with exactly one a because the initial variable generates exactly one a and no other rules can generate additional a 's. The variable A then generates zero or more b 's on both sides of the a .

- b. All strings with no more than three a 's.

$$S \rightarrow BABABAB$$

$$A \rightarrow a \mid \lambda$$

$$B \rightarrow bB \mid \lambda$$

This grammar can generate all strings with at most three a 's because the initial variable generates the required pattern. The variable A appears three times in the pattern produced by S , where each instance of A can produce exactly zero or one a . This ensures there are exactly zero, one, two, or three a 's in each string produced by the grammar. The variable B appears between each instance of the variable A and can generate zero or more b 's, thereby completing the full language.

In each case, give convincing arguments that the grammar you give indeed generate the indicated language.

3. Give a simple description of the language generated by the productions:

$$S \rightarrow aA$$

$$A \rightarrow bS$$

$$S \rightarrow \lambda$$

This language contains all strings with any number of ab pairs.

$$L = \{(ab)^n : n \geq 0\}$$

4. What language does the grammar with these productions?

$$S \rightarrow Aa$$

$$A \rightarrow B$$

$$B \rightarrow Aa$$

There is no production rule which ends in a terminal, so there are no finite strings produced by this grammar. Therefore, the language that is produced by this grammar is the empty set.

$$L = \{\emptyset\}$$

5. Find a grammar that generates the language

$$L = \{\omega\omega^R \mid \omega \in \{a,b\}^+\}$$

$$S \rightarrow aSa \mid bSb \mid aa \mid bb$$

6. Are the two grammars with respective productions

$$S \rightarrow aSb \mid ab \mid \lambda$$

and

$$S \rightarrow aAb \mid ab$$

$$A \rightarrow aAb \mid \lambda$$

equivalent? Assume that S is the start symbol in both cases.

The two grammars are not equivalent. The first grammar produces a language that includes λ while the second grammar produces a language that does not include λ .

7. So far we have given examples of only relatively simple grammars; every production had a single variable on the left side. As we will see, such grammars are very important, but the definition of grammar allows more general forms.

Consider the grammar $G = (\{A, B, C, D, E, S\}, \{a\}, S, P)$, with productions

$$\begin{aligned} S &\rightarrow ABaC \\ Ba &\rightarrow aaB \\ BC &\rightarrow DC|E \\ aD &\rightarrow Da \\ AD &\rightarrow AB \\ aE &\rightarrow Ea \\ AE &\rightarrow \lambda \end{aligned}$$

Derive three different sentences in $L(G)$. From these, make a conjecture about $L(G)$.

$$S \rightarrow ABaC \rightarrow AaaBC \rightarrow AaaE \rightarrow AaEa \rightarrow AEaa \rightarrow aa$$

$$S \rightarrow ABaC \rightarrow AaaBC \rightarrow AaaDC \rightarrow AaDaC \rightarrow ADaaC \rightarrow ABaaC \rightarrow AaaBaC \rightarrow AaaaaBC \\ \rightarrow AaaaaE \rightarrow AaaaEa \rightarrow AaaEaa \rightarrow AaEaaa \rightarrow AEaaaa \rightarrow aaaa$$

$$S \rightarrow ABaC \rightarrow AaaBC \rightarrow AaaDC \rightarrow AaDaC \rightarrow ADaaC \rightarrow ABaaC \rightarrow AaaBaC \rightarrow AaaaaBC \\ \rightarrow AaaaaDC \rightarrow AaaaDaC \rightarrow AaaDaaC \rightarrow AaDaaaC \rightarrow ADaaaaC \\ \rightarrow ABaaaaC \rightarrow AaaBaaaC \rightarrow AaaaaBaaC \rightarrow AaaaaaaBaC \\ \rightarrow AaaaaaaaaBC \rightarrow AaaaaaaaaE \rightarrow AaaaaaaaaEa \rightarrow AaaaaaaaaEaa \\ \rightarrow AaaaaaEaaa \rightarrow AaaaaEaaaa \rightarrow AaaaEaaaaa \rightarrow AaaEaaaaaa \\ \rightarrow AaEaaaaaaa \rightarrow AEaaaaaaaa \rightarrow aaaaaaaaa$$

$$S \rightarrow ABaC \rightarrow AaaBC \rightarrow AaaDC \rightarrow AaDaC \rightarrow ADaaC \rightarrow ABaaC \rightarrow AaaBaC \rightarrow AaaaaBC \\ \rightarrow AaaaaDC \rightarrow AaaaDaC \rightarrow AaaDaaC \rightarrow AaDaaaC \rightarrow ADaaaaC \\ \rightarrow ABaaaaC \rightarrow AaaBaaaC \rightarrow AaaaaBaaC \rightarrow AaaaaaaBaC \\ \rightarrow AaaaaaaaaBC \rightarrow AaaaaaaaaDC \rightarrow AaaaaaaaaDaC \rightarrow AaaaaaaaaDaaC \\ \rightarrow AaaaaaDaaaC \rightarrow AaaaaDaaaaC \rightarrow AaaaDaaaaaC \rightarrow AaaDaaaaaaC \\ \rightarrow AaDaaaaaaaC \rightarrow ADaaaaaaaaC \rightarrow ABaaaaaaaaC \rightarrow AaaBaaaaaaaaC \\ \rightarrow AaaaaBaaaaaaC \rightarrow AaaaaaBaaaaaC \rightarrow AaaaaaaaaBaaaaC \\ \rightarrow AaaaaaaaaaBaaC \rightarrow AaaaaaaaaaaaaaBaC \\ \rightarrow AaaaaaaaaaaaaaaaaBaC \rightarrow AaaaaaaaaaaaaaaaaaBC \\ \rightarrow AaaaaaaaaaaaaaaaaaE \rightarrow AaaaaaaaaaaaaaaaaaEa \\ \rightarrow AaaaaaaaaaaaaaaaaaEaa \rightarrow AaaaaaaaaaaaaaaaaaEaaa \\ \rightarrow AaaaaaaaaaaaaaaaaaEaaaa \rightarrow AaaaaaaaaaaaaaaaaaEaaaaa \\ \rightarrow AaaaaaaaaaaaaaaaaaEaaaaa \rightarrow AaaaaaaaaaaaaaaaaaEaaaaaa \\ \rightarrow AaaaaaaaaaaaaaaaaaEaaaaaa \rightarrow AaaaaaaaaaaaaaaaaaEaaaaaaa \\ \rightarrow AaaaaaEaaaaaaaaaaaa \rightarrow AaaaaEaaaaaaaaaaaaa \\ \rightarrow AaaaEaaaaaaaaaaaaa \rightarrow AaaEaaaaaaaaaaaaaa \\ \rightarrow AEaaaaaaaaaaaaaaaa \rightarrow aaaaaaaaaaaaaaaaa$$

The language contains all strings with a number of a 's equal to a power of two, starting at two.

$$L(G) = \{a^{2^n} : n > 0\}$$

8. Consider the grammar $G = (\{A, B, C, S\}, \{a, b, c\}, S, P)$, with productions

$$S \rightarrow abc$$

$$S \rightarrow A$$

$$A \rightarrow aABc \mid abc$$

$$cB \rightarrow Bc$$

$$bB \rightarrow bb$$

Derive three different sentences in $L(G)$. From these, make a conjecture about $L(G)$.

$$S \rightarrow abc$$

$$S \rightarrow A \rightarrow aABc \rightarrow aabcBc \rightarrow aabBcc \rightarrow aabbcc$$

$$\begin{aligned} S \rightarrow A \rightarrow aABc \rightarrow aaABcBc \rightarrow aaabcBcBc \rightarrow aaabBccBc \rightarrow aaabbccBc \\ \rightarrow aaabbcBcc \rightarrow aaabbBccc \rightarrow aaabbbccc \end{aligned}$$

The language contains all strings with an equal number of a 's, b 's, and c 's where all a 's precede all b 's that precede all c 's.

$$L(G) = \{a^n b^n c^n : n > 0\}$$
